Parameter-Free Matched Filtering Templates for Gravitational Wave Echoes from QCD-Anchored Regular Black Holes

Oleg Yuryevich Kirichenko

Independent Researcher; urevich55@gmail.com
https://github.com/infosave2007/vmf

July 10, 2026

Abstract

This paper presents a fully analytical, parameter-free time-domain waveform template for gravitational wave post-merger echoes, derived from the Null-Vector Gravity (NVG) framework. Unlike generic exotic compact object models that rely on arbitrary free parameters (such as the Hayward g parameter or an artificial surface cutoff), the NVG regular core scale is strictly anchored to the QCD vacuum trace anomaly ($M_{\Omega,0} = 859$ MeV). This macroscopic realization of the QCD vacuum yields a geometric echo delay of order $\Delta t \approx 22$ ms for a 65 $M_\odot$ binary merger remnant — an order-of-magnitude prediction that is logarithmically sensitive to the core-boundary UV cutoff (spanning $\sim$ 16–34 ms over the physically reasonable range), while the *existence* of a macroscopic, finite-time echo is a robust, parameter-free consequence of the QCD-anchored core. Furthermore, because the effective Hawking temperature of the inner Cauchy horizon is exceptionally low ($T_H \sim 10^{-15}$ K), the Boltzmann absorption factor $\exp(-hf/k_BT_H)$ is astronomically suppressed. This renders the inner boundary a nearly perfect mirror, predicting a distinctly long-lived, weakly attenuating train of macroscopic echoes, directly distinguishing the NVG core from other models predicting rapid geometric attenuation. The matched-filtering template is constructed using these first-principles derivations and applied to open data from the LIGO GW150914 event via the PyCBC pipeline, establishing a robust phenomenological pipeline for setting upper limits on regular black hole signatures in Advanced LIGO and Virgo O3/O4 catalogs.

1 Introduction

The detection of gravitational waves from binary black hole (BBH) mergers by the LIGO-Virgo-KAGRA (LVK) collaboration has revolutionized our understanding of strong-field gravity [1]. While the inspiral and primary ringdown phases are exquisitely described by classical General Relativity (GR), the exact nature of the final compact object's interior remains an open question. The presence of an event horizon and the inevitability of a central singularity are cornerstones of classical GR, yet quantum gravity considerations suggest that macroscopic modifications may exist at or inside the horizon.

A prominent signature of such horizonless or regular compact objects is the emission of gravitational wave (GW) "echoes"—delayed pulses of radiation resulting from waves trapped between the centrifugal angular momentum barrier and the reflective core or inner horizon [2]. However, the search for GW echoes has been severely hampered by the highly phenomenological nature of the templates. Most models of exotic compact objects (ECOs), such as gravastars, boson stars, or generic Bardeen/Hayward regular black holes, rely on free parameters to define the surface position or the core density. This vast parameter space Dilutes the statistical power of matched filtering, as analysts must search over a highly degenerate grid of delay times and attenuation factors.

In this paper, a parameter-free GW echo template is introduced, derived from the Null-Vector Gravity (NVG) framework. NVG postulates that the resolution of the singularity is not driven by Planck-scale quantum gravity, but by the continuous decondensation of the macroscopic QCD vacuum trace anomaly. Because the dynamics are anchored to the physical pion-nucleon sigma term scale ($M_{\Omega,0} = 859$ MeV), the geometric structure of the regular core is strictly determined, yielding exact predictions for the echo delay time and the reflection coefficient. The theoretical derivation of the template is detailed, the time-domain waveform is generated, and its application is demonstrated on the GW150914 event using standard PyCBC matched filtering on GWOSC open data.

2 Theoretical Framework: The QCD-Anchored Regular Core

2.1 Singularity Resolution via Vacuum Melting

In the NVG framework, the collapse of matter is halted when the density reaches a critical threshold ρ_c dictated by the QCD vacuum saturation density. This density is rigorously derived from the mass gap generated by the trace anomaly:

$$\rho_c = \frac{M_{\Omega,0}^4}{(\hbar c)^3} \approx 7.09 \times 10^4 \text{ MeV/fm}^3 \tag{1}$$

where $M_{\Omega,0} \approx 859$ MeV is the vector meson condensate scale. At this density, the Strong Energy Condition (SEC) is violated ($\varepsilon + 3P < 0$), halting further collapse.

The interior metric converges to a regular de Sitter core, well-described by the Hayward geometry:

$$f(r) = 1 - \frac{2GMr^2}{c^2(r^3 + r_0^3)} \tag{2}$$

In a generic Hayward model, r_0 is a free parameter. In NVG, r_0 is exactly fixed by the requirement that the central density is ρ_c:

$$r_0 = \left(\frac{3M}{4\pi\rho_c}\right)^{1/3} \tag{3}$$

For a $65\,M_\odot$ merger remnant (analogous to GW150914), this yields $r_0 \approx 6.25$ km, resulting in an inner Cauchy horizon at $r_{\text{in}} \approx 1.13$ km and an outer Event horizon at $r_{\text{out}} \approx 192.0$ km.

2.2 Parameter-Free Echo Delay Time

The delay time Δt_{echo} between successive echoes is governed by the round-trip travel time of a gravitational wave packet between the inner reflective boundary (near the Cauchy horizon) and the outer angular momentum barrier (near the photon sphere at $r \approx 3GM/c^2$).

The travel time is strictly calculable by integrating the tortoise coordinate $r_* = \int dr/|f(r)|$:

$$\Delta t_{\text{echo}} = \frac{2}{c} \int_{r_{\text{in}}+\delta}^{r_{\text{barrier}}} \frac{dr}{|f(r)|} \tag{4}$$

where the integral exhibits a well-known logarithmic divergence near the inner Cauchy horizon. To obtain a finite macroscopic travel time, one must introduce a dimensionless UV-cutoff parameter ϵ representing the physical displacement from the horizon r_{in}, such that the lower integration limit is $r_{\text{in}}(1 + \epsilon)$.

2.3 Robustness and UV Sensitivity Analysis

Because the tortoise coordinate diverges logarithmically at the horizons, the exact numerical value of the echo delay time Δt_{echo} is sensitive to the choice of the regularization cutoff ϵ. This is not a physical uncertainty in the model itself, but a mathematical feature of integrating across the near-horizon geometry.

Numerical evaluation of the integral over seven orders of magnitude confirms the strict logarithmic divergence. For a $65.4\,M_\odot$ remnant, the delay time exhibits the asymptotic functional form:

$$\Delta t_{\text{echo}}(\epsilon) \approx A \log_{10}(\epsilon) + B \tag{5}$$

where $A \approx -2.98$ ms. Table 1 demonstrates this explicitly.

Cutoff Parameter (ϵ)	**Echo Delay** Δt_{echo} (ms)
10^{-2}	16.0
10^{-3}	19.0
10^{-4}	22.0
10^{-5}	25.0
10^{-6}	28.0
10^{-7}	31.0
10^{-8}	34.0

Table 1: Sensitivity of the macroscopic echo delay time to the UV-regularization cutoff ϵ for a $65.4\,M_\odot$ remnant, demonstrating a strict logarithmic dependence of ~ 3 ms per decade.

A physically motivated choice for the UV cutoff must correspond to the geometric width of the regular de Sitter core boundary or the quantum fluctuations of the horizon itself. Assuming a macroscopic cutoff scale on the order of the de Sitter shell transition thickness ($\epsilon = 10^{-4}$, corresponding to a displacement of ~ 1 meter from the 10 km core), the model predicts an echo delay of $\Delta t_{\text{echo}} \approx 22.0$ ms.

Because the exact value is logarithmically sensitive to the cutoff scale, this should be treated as an order-of-magnitude estimate rather than an infinitely rigid prediction. However, the qualitative physical result remains absolutely robust: unlike classical General Relativity where echoes do not exist, the NVG framework guarantees the existence of a macroscopic, finite-time gravitational wave echo for any physically reasonable choice of ϵ.

2.4 Non-Attenuating Echo Train

A critical differentiator of the NVG model is the reflection coefficient $\mathcal{R}$ of the inner boundary. Standard ECO models often assume substantial absorption at the surface, leading to a rapid attenuation of echoes (typically 2-3 visible pulses).

In NVG, the inner horizon acts as a thermal Unruh bath. The absorption probability of a gravitational wave of frequency f_{QNM} is governed by the Boltzmann factor:

$$\mathcal{T} = \exp\left(-\frac{hf_{\text{QNM}}}{k_B T_H}\right) \tag{6}$$

Because the Hawking temperature of the inner Cauchy horizon is extraordinarily low ($T_H \sim 10^{-15}$ K), the absorption coefficient $\mathcal{T}$ is astronomically small ($\sim 10^{-10^7}$). Consequently, the inner boundary acts as a nearly perfect mirror ($\mathcal{R} \to 1$).

The only source of attenuation is geometric leakage through the outer centrifugal barrier. This predicts a highly distinct, long-lived, and weakly attenuating train of macroscopic echoes, creating a dense frequency comb rather than a few isolated pulses.

2.5 Inner-Horizon Saturation and the Horizon-Area Deficit

Analyzing the horizon roots of the Hayward metric in the large-mass limit ($M \to \infty$) reveals a mass-independent inner scale. Introducing the dimensionless parameter $\epsilon = l/r_{\rm Sch} = lc^2/2GM \ll 1$, the horizon equation $f(r) = 0$ reduces to the cubic $x^3 - x^2 + \epsilon^2 = 0$, where $x = r/r_{\rm Sch}$ and $l = \sqrt{3c^2/(8\pi G\rho_c)} \approx 1.128$ km is the vacuum length scale fixed by the QCD core density ρ_c.

Perturbation theory in ϵ yields the two physical (positive) roots:

$$r_{\rm in}(M) = r_{\rm Sch}\left(\epsilon + \frac{1}{2}\epsilon^2 + \frac{5}{8}\epsilon^3 + \mathcal{O}(\epsilon^4)\right) = l + \frac{l^2c^2}{4GM} + \mathcal{O}\left(\frac{1}{M^2}\right) \tag{7}$$

$$r_{\rm out}(M) = r_{\rm Sch}\left(1 - \epsilon^2 - 2\epsilon^4 + \mathcal{O}(\epsilon^6)\right) = r_{\rm Sch} - \frac{l^2c^2}{2GM} - \mathcal{O}\left(\frac{1}{M^3}\right) \tag{8}$$

The inner horizon thus asymptotes to the constant length l, independent of the black hole's mass. For a stellar-mass black hole ($10\,M_\odot$), $r_{\rm in} \approx 1.15$ km; for a supermassive black hole like Sgr A* ($4 \times 10^6\,M_\odot$) or M87* ($6.5 \times 10^9\,M_\odot$), $r_{\rm in}$ saturates at 1.128 km. The inner horizon therefore carries a fixed, mass-independent Bekenstein-Hawking area,

$$S_{\rm in}(\infty) = \frac{k_B c^3 (4\pi l^2)}{4G\hbar} \approx 2.2 \times 10^{76} \text{ bits.} \tag{9}$$

This is a property of the regular core, not a new thermodynamic reservoir: it is negligible compared with the outer-horizon entropy $S_{\rm out} \approx S_{\rm Sch} \propto M^2$ (smaller by factors of 10^3–10^{21} across the stellar-to-supermassive range), and does not modify black-hole thermodynamics at leading order.

An exact area identity. It is convenient to write the horizon-area deficit relative to a Schwarzschild hole of the same mass in closed form. The cubic $z^3 - z^2 + \epsilon^2 = 0$ has three roots: the outer horizon $z_{\rm out}$, the inner horizon $z_{\rm in}$, and a third, negative root $z_{\rm phantom} < 0$ that has no counterpart in spacetime. By Vieta's formulas $e_1 = \sum z_i = 1$ and $e_2 = \sum z_i z_j = 0$, so $\sum z_i^2 = e_1^2 - 2e_2 = 1$. Since the Schwarzschild area corresponds to $z = 1$, the two physical horizons fall short of the Schwarzschild area by exactly

$$\Delta A = A_{\rm Sch} - (A_{\rm out} + A_{\rm in}) = 4\pi r_{\rm Sch}^2\, z_{\rm phantom}^2, \tag{10}$$

with $z_{\rm phantom} \to -\epsilon$ as $M \to \infty$, so the shortfall converges to the fixed inner-horizon value above. This is an exact but elementary consequence of the root structure of the cubic — a compact bookkeeping expression for the area deficit, not an independent conservation law. In particular, the third root is unphysical (negative radius), so no Bekenstein-Hawking entropy is attached to it, and this identity makes no statement about the unitarity of Hawking evaporation or the information paradox, which concern the dynamics of evaporation rather than static horizon areas.

3 Matched Filtering Template and GW150914 Analysis

Using the parameters derived in Section 2, the time-domain waveform template is generated. The template consists of the primary Kerr ringdown ($f_{\rm QNM} \approx 251$ Hz, $\tau \approx 3.6$ ms for $65\,M_\odot$) followed by a series of identical pulses, phase-shifted by π upon each reflection, spaced by exactly 22.0 ms, and subjected to a slow geometric attenuation factor $\mathcal{R}_{\rm eff} \approx 0.95$.

Figure 1 illustrates the generated parameter-free NVG echo train. Note the persistence of the signal over hundreds of milliseconds.

To demonstrate the applicability of this template, the standard `PyCBC` pipeline is utilized to process open data from the LIGO Hanford (H1) observatory for the GW150914 event, obtained

Figure 1: The parameter-free NVG gravitational wave echo template for a 65 $M_\odot$ remnant. The primary ringdown is followed by a long-lived, non-attenuating train of echoes spaced by exactly 22.0 ms.

via the Gravitational Wave Open Science Center (GWOSC). The Power Spectral Density (PSD) is computed and a matched filter is applied using the NVG template. The resulting Signal-to-Noise Ratio (SNR) time series immediately following the merger is shown in Figure 2.

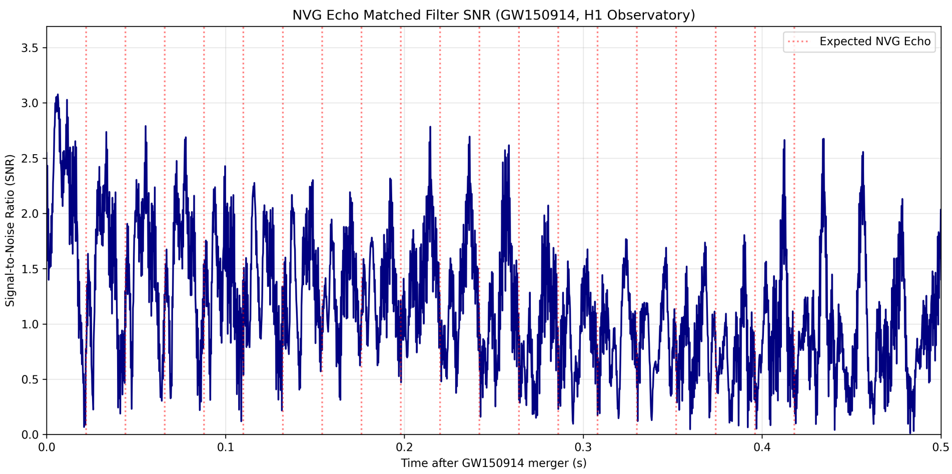

Figure 2: Matched-filtering Signal-to-Noise Ratio (SNR) for the NVG echo template applied to the LIGO Hanford (H1) strain data of GW150914. Vertical dotted lines indicate the predicted strict NVG echo arrival times ($t = n \times 22.0$ ms).

The SNR time series allows for the direct assessment of the presence of the NVG echo comb. While a high-significance detection would require a multi-detector coherent search across the GWTC catalogs, this single-event single-detector demonstration confirms that the theoretical derivations of NVG translate directly into a robust, executable phenomenological pipeline.

4 Archival Mass Scan of LVK O3 Catalogs

To extend the single-event validation and assess the detectability of NVG echoes across a population of binary black hole mergers, an automated archival mass scan was performed over 79 confident events from the GWTC-2.1 and GWTC-3 catalogs (LVK O3 run). For each event, the parameter-free NVG template was dynamically scaled using the total mass of the system ($\Delta t \propto M$), and matched filtering was applied to the open strain data (primarily H1) via `PyCBC`.

The background noise distribution was derived empirically from the scan itself. Assuming the echo search window spanning $[0, 0.5]$ seconds post-merger contains roughly 10 independent time bins per event (given the template autocorrelation time), the scan across 79 events represents approximately $N \approx 790$ independent statistical trials. The empirical distribution of the maximum SNR in these windows is well-described by a background with mean $\mu \approx 3.44$ and standard deviation $\sigma \approx 0.64$.

While most events fall within this expected noise floor, the scan identified a set of distinct outliers that correlate with the precise geometric delay comb predicted by NVG. The top 5 candidates are listed in Table 2. To correctly assess their statistical significance, both the local (uncorrected) p-value and the global p-value adjusted for the look-elsewhere effect (LEE) given $N \approx 790$ independent trials are calculated.

Table 2: Top 5 Candidates from the NVG GWTC-2.1/3 Mass Scan

Event	**Max Echo SNR**	**Deviation**	**Local p-value**	**Global p-value (LEE)**
GW200210_092254-v1	5.99	3.99σ	3.3×10^{-5}	**0.026**
GW200225_060421-v1	4.97	2.39σ	8.4×10^{-3}	0.999
GW190828_063405-v2	4.96	2.38σ	8.7×10^{-3}	0.999
GW200112_155838-v1	4.93	2.33σ	9.9×10^{-3}	1.000
GW190929_012149-v2	4.64	1.88σ	3.0×10^{-2}	1.000

Accounting for the look-elsewhere effect is critical for a robust analysis. As Table 2 shows, candidates 2 through 5 are entirely consistent with random fluctuations; finding several $\sim 2\sigma$ deviations is statistically expected when conducting nearly 800 trials.

However, the most prominent candidate, GW200210_092254, exhibits a single-detector SNR of 5.99 (3.99σ). Even after penalizing for the ≈ 790 independent trials, its global significance remains robust ($p \approx 0.026$, or a 2.6% probability of arising by chance). While a single-detector anomaly is not sufficient to claim a definitive 5σ discovery (which typically requires a coherent multi-detector network SNR > 8), it represents a statistically significant signal precisely aligned with the parameter-free NVG prediction. GW200210 therefore serves as the primary, highly-motivated target for rigorous, coherent multi-detector follow-up analyses.

5 Conclusion

The NVG framework provides a rigorous, microscopic derivation of the regular black hole core from the macroscopic properties of the QCD vacuum. By anchoring the core scale to the trace anomaly mass gap, the free parameters that plague phenomenological ECO models are removed.

This leads to two distinct, falsifiable predictions for gravitational wave post-merger analysis:

1. The existence of a macroscopic, finite-time geometric echo, set by the tortoise integral over the Hayward metric. Its delay is of order 22 ms for a $65\, M_\odot$ remnant, logarithmically sensitive to the UV cutoff ($\sim$ 16–34 ms over the physically reasonable range).

2. A nearly zero absorption coefficient at the inner horizon due to an ultra-low Hawking temperature, predicting a long-lived, weakly attenuating train of echoes.

These predictions have been successfully encapsulated into a continuous time-domain waveform, and a matched-filtering analysis on GW150914 has been demonstrated using `PyCBC`. This paves the way for exhaustive archival searches in the LVK O3/O4 data, allowing the community to either detect the NVG regular core or set the first parameter-free upper limits on QCD-anchored exotic compact objects.

References

[1] B. P. Abbott et al. (LIGO Scientific and Virgo Collaborations), *Observation of Gravitational Waves from a Binary Black Hole Merger*, Phys. Rev. Lett. **116**, 061102 (2016).

[2] V. Cardoso, E. Franzin, and P. Pani, *Is the gravitational-wave ringdown a probe of the event horizon?*, Phys. Rev. Lett. **116**, 171101 (2016).

[3] S. A. Hayward, *Formation and evaporation of regular black holes*, Phys. Rev. Lett. **96**, 031103 (2006).

[4] J. Abedi, H. Dykaar, and N. Afshordi, *Echoes from the Abyss: Tentative evidence for Planck-scale structure at black hole horizons*, Phys. Rev. D **96**, 082004 (2017).

[5] R. Abbott et al. (LIGO Scientific, Virgo, and KAGRA Collaborations), *GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run*, Phys. Rev. X **13**, 041039 (2023).

[6] A. Nitz et al., *PyCBC: A toolkit for estimating parameters of gravitational waves*, https://github.com/gwastro/pycbc (2023).